

Project Proposal (Proposed Solution) Report

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Title	AI StudyBuddy — AI-Powered Study Assistant for Smarter Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposal aims to transform study preparation using Generative AI, boosting efficiency and saving time for students. It tackles the challenges of manual note-making, flashcard creation, and unstructured study by providing AI-automated learning tools. Key features include AI-powered summarization, flashcard generation, quiz generation, and personalized study planning via Google Gemini.

Project Overview

Field	Details
Objective	The primary objective is to build an AI-powered backend API that automates study preparation tasks — generating summaries, flashcards, quizzes, and personalized study plans from uploaded study materials using Google Gemini Generative AI.
Scope	The project comprehensively covers user authentication (JWT), study material management (upload/CRUD), AI content generation (4 features via Gemini API), admin management, and data persistence using MongoDB Atlas. It is a backend-only REST API project tested via Postman.

Problem Statement

Field	Details
Description	Students struggle with lengthy study materials, spend excessive time on manual note-making and flashcard creation, lack self-assessment tools, have no personalized study plans, and use multiple disconnected tools for study preparation.
Impact	Solving these issues results in significant time savings, better comprehension through AI-generated summaries, effective revision through flashcards, self-assessment through quizzes, organized study through personalized plans, and a single centralized platform for all learning resources.

Proposed Solution

Field	Details
Approach	Employing Google Gemini 2.5 Flash Generative AI API to analyze uploaded study material content and generate learning resources. The backend is built with Node.js + Express.js following MVC architecture, with MongoDB Atlas for data persistence and JWT for secure authentication.
Key Features	• JWT-based user authentication with bcrypt password hashing and refresh tokens. • File upload support (.txt, .md, .pdf) up to 5MB via Multer. • AI Summary generation (bullet-point format). • AI Flashcard generation (JSON Q&A pairs). • AI MCQ Quiz generation (JSON with options and correct answers). • AI Personalized Study Plan (day-by-day schedule based on goal, hours, and days). • Admin panel (user management)

Problem-Solution Fit Canvas

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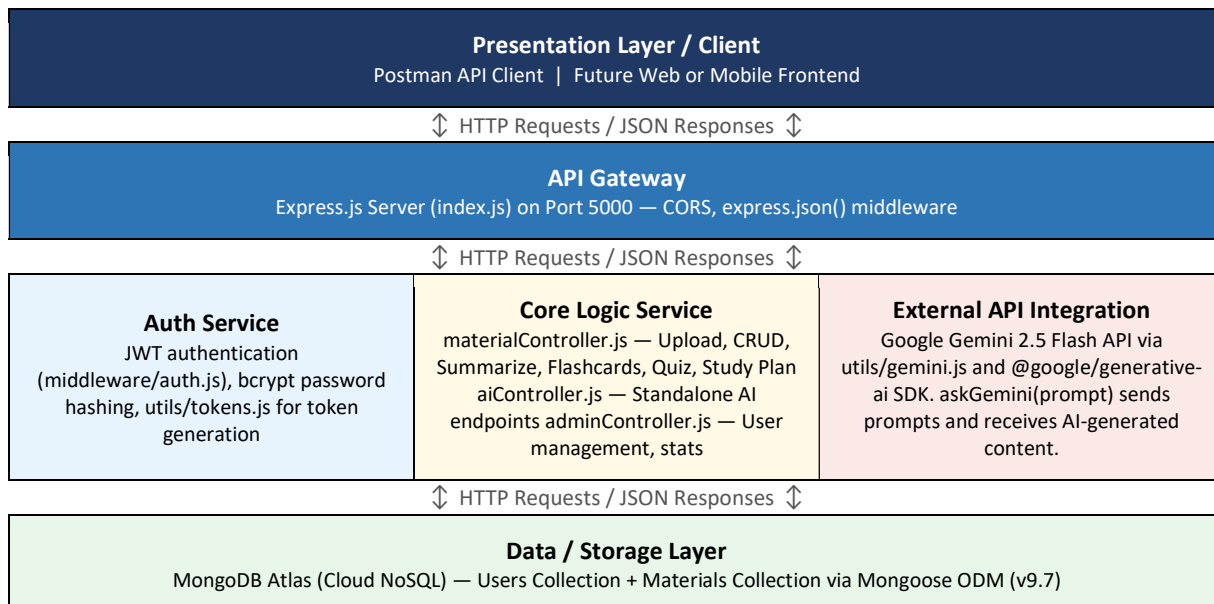
Field	Details
1. CUSTOMER SEGMENT(S) [CS]	College students preparing for semester exams. School students preparing for board exams. Competitive exam aspirants. Self-learners studying independently.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]	Difficulty understanding lengthy study materials (Daily). Manual note-making and flashcard creation is time-consuming (Daily). No self-assessment tools from own material (Weekly). Lack of personalized study plans (Every exam cycle). Scattered study resources across multiple apps (Daily).
3. TRIGGERS TO ACT [TR]	Approaching exam deadlines. Failing to retain information from long reading sessions. Seeing classmates use AI tools for study preparation. Frustration with manual flashcard creation.
4. EMOTIONS BEFORE / AFTER [EM]	BEFORE: Overwhelmed, anxious, frustrated, directionless, stressed. AFTER: Confident, relieved, organized, empowered, prepared.
5. AVAILABLE SOLUTIONS PROS & CONS [AS]	ChatGPT/Gemini directly: Pro — Free AI access. Con — No file upload, no persistence, no authentication, no structured output. Quizlet: Pro — Flashcard creation. Con — Manual input required, no AI generation from own material. YouTube summaries: Pro — Visual explanations. Con — Not personalized to student's own material.
6. CUSTOMER LIMITATIONS [CL]	Limited budget (students prefer free tools). Need internet access for AI API calls. Limited time during exam preparation periods. Basic tech proficiency (need simple API interface).
7. BEHAVIOR + ITS INTENSITY [BE]	Reading long chapters and highlighting manually (High intensity, daily). Procrastinating on difficult subjects due to lack of structure (Medium intensity). Cramming in the last few days before exams (High intensity, every exam). Searching for shortcuts and online summaries (Medium intensity).
8. CHANNELS OF BEHAVIOR [CH]	ONLINE: Postman API client, REST API endpoints, MongoDB Atlas cloud database, Google Gemini API. OFFLINE: Study material files (.txt, .md, .pdf), handwritten notes converted to text files.
9. PROBLEM ROOT CAUSE [RC]	The volume of study material exceeds the student's available time for manual processing. No existing tool combines AI summarization + flashcards + quizzes + study planning from uploaded content in one authenticated platform. Students lack expertise to create quality self-assessment materials.
10. YOUR SOLUTION [SL]	AI StudyBuddy — A RESTful backend API built with Node.js, Express, and MongoDB that allows authenticated students to upload study materials and uses Google Gemini AI to automatically generate: (1) Concise summaries, (2) Q&A flashcards, (3) MCQ quizzes, (4) Personalized study plans. All generated content is persisted in MongoDB and accessible anytime via secure JWT-authenticated API endpoints.

Solution Architecture

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Solution Architecture Diagram

A layered architecture diagram showing how data and requests flow through the AI StudyBuddy system from client to database.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	API client for sending HTTP requests and receiving JSON responses. Currently Postman is used for testing and demonstration. Future frontend (React/Next.js) can consume the same API.	Postman, HTTP/REST
API Gateway	Express.js server (index.js) that receives all incoming requests, applies middleware (CORS, JSON parsing), and routes to appropriate controllers. Listens on Port 5000.	Express.js v4.22, cors, express.json()
Auth Service	Handles user registration, login, token refresh, and logout. Passwords hashed with bcrypt. JWT access tokens (15 min) and refresh tokens (7 days) generated via utils/tokens.js. Role-Based Access Control via middleware (protect + adminOnly).	jsonwebtoken v9.0, bcryptjs v3.0, middleware/auth.js
Core Logic Service (Controllers)	materialController: File upload via Multer, CRUD operations, AI content generation (summarize, flashcards, quiz, study-plan). aiController: Standalone AI endpoints accepting materialId in body. adminController: List users, delete users, platform statistics.	Express route handlers, Multer v2.2, Mongoose queries
External API Integration	Google Gemini 2.5 Flash API integration via utils/gemini.js. The askGemini(prompt) function sends constructed prompts to Gemini and returns AI-generated text. Used for all four AI features.	@google/generative-ai SDK v0.24, Google Gemini 2.5 Flash model
Data / Storage Layer	MongoDB Atlas cloud database with two collections: Users (name, email, hashedPassword, role) and Materials (userId, title, content, summary, flashcards[], quiz[], studyPlan). Mongoose ODM provides schemas, validation, and query interface.	MongoDB Atlas, Mongoose v9.7, User.js and Material.js schemas